Breaking the Lexical Supervision Ceiling: Clause-Level MiniLM-L6 Verification and Task-Priority Intent Routing on a Zero-VRAM Edge NPU

Sean Maxwell
Independent Researcher · ORCID: 0009-0004-0133-0463
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Abstract. In a prior study (DOI: 10.5281/zenodo.23901605), we showed that deterministic speech-regex and stem-list action supervisors reach an empirical ceiling on live language model outputs: across *N=1{,}000* four-model completions, a 44-stem grammar supervisor caught *71.1%* (*128/180*) of in-vocabulary action claims but dropped to *29.5%* (*23/78*) on unseen verb lemmas (*58.5%* overall), while Structured-Output Action Contracts (SOAC) triggered spurious action frames on *13.3%–16.7%* of conversational idioms. Replacing lexical rules with a neural classifier solves open-vocabulary generalization on isolated single sentences, yet introduces two new failure modes in edge companion deployments: (1) dGPU VRAM contention with the primary language model, and (2) severe probability dilution when a single fabricated action clause or task request is buried inside a multi-sentence response. In this paper, we present a three-classifier neurosymbolic supervision and routing architecture (`cross-encoder/nli-MiniLM2-L6-H768`, *82.1M* parameters) compiled to FP16 on an Intel AI Boost NPU (*0 MB* dGPU VRAM, *1.22–2.95 ms* mean inference latency on an Intel Core Ultra 9 275HX). By decomposing multi-sentence inputs into syntactic clauses (`split_narration_into_clauses`) and applying domain-specific pooling and symbolic verification operators—clause-level speech-act classification paired with deterministic receipt set-membership verification for action claims, max-hazard lattice pooling for epistemic proof policy, and task-priority pooling for multi-head intent routing—we eliminate multi-sentence dilution without retraining the frozen OpenVINO FP16 NPU weights. Across an expanded *N=2{,}022* action-claim evaluation battery (combining the 1,000-turn 4-model corpus, 715-turn Paper 02 v1/v2 6-arm corpus, 300 organic companion discovery/held-out turns, and 30 implicit zero-verb stress probes), the clause-level NPU verifier achieves **99.26% accuracy (*2{,}007/2{,}022*, 95% Wilson CI: *[98.78%, 99.55%]*)**, **98.73% recall (*465/471*)**, and **0.58% FPR (*9/1{,}551*)**, outperforming the 44-stem regex baseline (*11.46%* recall [*54/471*], McNemar $\chi^2 = 399.04$, $p = 3.07 \times 10^{-89}$). On a pre-registered 150-item code-and-weight-locked holdout (`WO42_LOCKED_HOLDOUT_150`) authored by three isolated subagents and graded by blinded raters ($\kappa = 1.000$), the three NPU classifiers achieve *145/150* (*96.67%*, 95% CI: *[92.43%, 98.57%]*) single-pass accuracy and *150/150* (*100.0%*) after fixing two deterministic router rules (`has_z3` domain precedence and Markdown backtick stripping around app names). Across a live 100-turn end-to-end evaluation spanning all four local Gemma-4 GGUF checkpoints (`26B-A4B`, `12B`, `E4B`, `E2B`), the system achieves **100/100 (100.0%)** functional coherence and grounding and **98/100 (98.0%)** task/reasoning completion. Finally, to resolve the sample-size limits of our earlier preprints (DOI: 10.5281/zenodo.23901452, DOI: 10.5281/zenodo.23901683), we execute pre-registered *N=300* expansions (`prereg-wo43-power-300`) of the 5-arm Memory Poisoning Ablation (*0/100* [*0.0%*, 95% CI: *[0.00%, 4.61%]*] Cohort A self-poisoning and Cohort C user-misinformation absorption vs. *100/100* Naive RAG, exact McNemar $p = 4.16 \times 10^{-23}$) and the 3-arm Contrastive Companion Battery (**100/100** Seeded Recall, **100/100** Receipt Introspection, and **100/100** Calibrated Abstention on Compound Lucca, outperforming Bare LLM by *+98.0 pp* on Recall [$p_{HB} = 1.03 \times 10^{-21}$] and *+50.0 pp* on Receipt Introspection

[$p_{HB} = 2.95 \times 10^{-11}$], and outperforming Top-5 Hybrid RAG by $+15.0$ pp on Recall [$p_{HB} = 3.05 \times 10^{-4}$]).

Introduction

An autonomous local agent that executes code, modifies workspace files, and queries external networks must satisfy two distinct honesty invariants. First, it must not poison its own long-term memory by storing unverified assistant narration as grounded fact. In our first study [maxwell2026poisoning] (DOI: 10.5281/zenodo.23901452) and our five-layer memory report [maxwell2026fivelayer] (DOI: 10.5281/zenodo.23901683), we enforced this storage invariant through six HKDF-derived HMAC-SHA256 signers and SQLite triggers that partition autobiographical recollections (`memory_eligible=1, grounding_eligible=0`) from verified facts (`grounding_eligible=1`).

Second, the agent must not claim in live speech or text that it has executed a state-modifying action when no corresponding `ToolReceipt` exists in the current turn. In our second study [maxwell2026evasion] (DOI: 10.5281/zenodo.23901605), we evaluated two deterministic mechanisms for enforcing this runtime output invariant:

1. \textbf{Lexical Speech-Regex Supervision (`ActionClaimGrammarSupervisor`):} Expanding a first-person past-tense regex from 21 verbs to 44 stems and adding progressive participle rules raised in-vocabulary sensitivity to *71.1%* (*128/180*) across *N=1{,}000* completions from four Gemma-4 models. However, sensitivity collapsed to **29.5% (23/78)** on 10 unseen out-of-dev action verbs (*58.5%* [*151/258*] overall).
2. **Structured-Output Action Contracts (SOAC):** Requiring typed `[ACTION: ...]` frames verified against cryptographic receipts eliminated unframed prose leakage (*0/120* on `gemma-4-26B-A4B`), but the contract system prompt caused the model to emit spurious action frames on *13.3%* (*4/30*) to *16.7%* (*5/30*) of figurative conversational idioms (e.g., *"kill two birds with one stone"*).

Paper 02 [maxwell2026evasion] therefore left an open architectural question: how can an edge agent verify open-vocabulary action claims, classify epistemic proof-policy hazards, and route multi-sentence user intents at *<5 ms* latency without consuming dGPU VRAM or failing on multi-sentence prose?

When we trained *82.1M*-parameter `cross-encoder/nli-MiniLM2-L6-H768` sequence classifiers on our discrete GPU and compiled them to OpenVINO FP16 on the Intel AI Boost NPU (`WO-41`), single-sentence in-distribution accuracy exceeded *98%*, but on the 90-item `WO-41` blind adversarial stress test accuracy dropped to *76.7%* (*69/90*), and on an uncurated 300-item multi-sentence blind benchmark (`Blind-300`, `WO-42 Phase 1`) pre-pooling accuracy dropped to **75.0% (225/300)**: *83.0%* (*83/100*) on action-claim verification, *69.0%* (*69/100*) on hazard classification, and *73.0%* (*73/100*) on intent routing (*303/390 = 77.7%* across the combined 390 blind items).

This paper diagnoses the root cause of that multi-sentence degradation and presents the clause-level neurosymbolic pooling architecture that resolves it.

Why Whole-Turn Neural Classifiers Degrade on Multi-Sentence Turns

Line-by-line inspection of the 87 failures in the pre-pooling `Blind-390` run (`logs/test_reports/wo42_pass1_pre_pooling_fix_report.json`) identified three deterministic failure mechanisms when transformer sequence classifiers score an entire multi-sentence turn as a single token span or attempt discrete set-membership in embedding space:

Mechanism 1: Neural Set-Membership Confusion and Multi-Sentence Dilution (*21/130* Coherence Misses)

Feeding `verified_tools` as a text string into a continuous embedding classifier forced the network to approximate discrete set membership (*claimed_tool* ∈ *executed_tools*) and caused sensitivity to argument formatting (`launch_app(app='...')` vs. `["launch_app"]`). Furthermore, when an assistant response contained 3 to 5 sentences where a grounded first clause (e.g., `read_file`) was followed by an ungrounded second clause (e.g., `write_file`), whole-paragraph encoding diluted the ungrounded second clause.

Mechanism 2: Benign Context Masking High-Hazard Clauses (*32/130* Hazard Misses)

In the epistemic proof-policy gate (`src/lucca/core/proof_policy.py`), a prompt containing three casual conversational sentences followed by a single factual/historical verification or destructive security question was classified as `LOW` hazard when scored as a single paragraph, while imperative non-corroboration commands falsely triggered `CROSS_REFERENCE_CMD`.

Mechanism 3: Conversational Opener Hijacking and Cross-Clause Entity Bleed (*34/130* Router Misses)

Natural voice turns frequently open with greetings, venting, or context before stating a concrete task. Whole-utterance classification averaged 3 to 6 sentences into low-confidence predictions or delegated exact lexical contracts (`EMERGENCY_REFLEX` and whitelisted `known_apps` vs. third-party apps) to unconstrained softmax heads without slot verification.

Clause-Level Neurosymbolic NPU Architecture

Rather than retraining larger models that would exceed our *<5 ms* latency budget, we keep all three *82.1M*-parameter `nli-MiniLM2-L6-H768` OpenVINO FP16 models pinned to the Intel AI Boost NPU (*0 MB* dGPU VRAM) and introduce a deterministic symbolic decomposition, receipt set-membership, and priority-pooling layer around each classifier.

\subsection{Syntactic Clause Segmentation (`split_narration_into_clauses`)} Given an input text *T*, `split_narration_into_clauses(T)` decomposes *T* into an ordered sequence of clauses *C = (c_1, c_2, ..., c_m)* using sentence punctuation boundaries and coordinating action conjunctions (`and I also`, `then I`, `;`) while protecting decimal numbers, file extensions, abbreviations, and fenced code blocks.

\subsection{Clause-Level Speech-Act Classification and Deterministic Receipt Set-Membership (`NeuralActionClaimVerifier`)} For assistant output *T_a* and user prompt *T_u*, `NeuralActionClaimVerifier` (`src/lucca/sandbox/neural_claim_verifier.py` [*1{,}082 LOC*], `models/action_claim_verifier_ov`, static shape `[1, 128]`) classifies each clause *c_i* ∈ *split_narration_into_clauses(T_a)* in the context of the current user turn (`format_pair_texts` isolated to `user` role messages) into 5 speech-act classes and 7 canonical tool families (`CLAIM_WRITE_NOTE`, `CLAIM_SEARCH_READ`, `CLAIM_EXEC_MATH`, `CLAIM_SCHEDULE`, `CLAIM_MEMORY`,

`CLAIM_LAUNCH_EXEC`, `CLAIM_DELETE_MOD`). Whenever a clause predicts an action claim or live-state fabrication, `verify_clause_claims_against_receipts` (`src/lucca/core/coherence.py`) performs exact deterministic Python set-membership and verb-subfamily verification against the turn's actual `verified_tools` (`ok=True`) receipts:

$$Breach(T_u, T_a, \mathcal{R}_{ok}) = \exists\, c_i \in clauses(T_a) : ClaimedFamily(T_u, c_i) \neq \emptyset \;\wedge\; ClaimedFamily(T_u, c_i) \cap Families(\mathcal{R}_{ok}) = \emptyset$$

\subsection{Max-Hazard Lattice Pooling, Legal Standards of Proof, and Walton Defeasible Corroboration (`NeuralHazardClassifier`)} For user prompt *T_u*, `NeuralHazardClassifier` (`src/lucca/core/neural_hazard_classifier.py`, `models/proof_policy_hazard_ov`, static shape `[1, 96]`) scores each sentence across 5 hazard classes (`LOW_BENIGN`, `LOW_CAPABILITY_OR_META`, `HIGH_FACTUAL_HISTORICAL`, `CROSS_REFERENCE_CMD`, `HIGH_DESTRUCTIVE_SECURITY`) and applies max-hazard sentence pooling, a `_HAS_CORROBORATION_SIGNAL_RE` necessity gate on `CROSS_REFERENCE_CMD`, and high-entropy/gibberish fail-closed escalation to `HIGH`.

The resulting hazard level $h \in \{LOW, MEDIUM, HIGH\}$ parameterizes a two-dimensional decision-theoretic burden-of-proof table *POLICY_TABLE_2D(c, h)* in `src/lucca/core/proof_policy.py` (`Spec D-REAS-09`) across four belief consumers $c \in \{casual_mention, memory_commit_inferred, unhedged_assertion, irreversible_action\}$ and four legal standards of proof (`ProofStandard` in `src/lucca/core/schemas.py`):

- \textbf{`SPECULATION` (*confidence* < *0.50*, Unlikely / Speculative):} Cannot be committed to memory or stated as belief; if voiced at conclusion of an abductive chain (`DeductionAndPlanningRunner.run_deduction_step` in `src/lucca/core/deduction_runner.py`), it requires an explicit speculation hedge (*"I'm guessing ..., but that's just speculation"*).
- \textbf{`BALANCE_OF_PROBABILITIES` (*0.50* ≤ *confidence* < *0.75*, More Likely Than Not):} Satisfies `(casual_mention, LOW)` with a hedged belief marker (*"I think ..., but I'm not sure enough to keep it"*), but is rejected by `meets_standard(confidence, "memory_commit_inferred")`.
- \textbf{`CLEAR_AND_CONVINCING` (*0.75* ≤ *confidence* < *0.95*):} Required minimum threshold to commit an `INFERRED` fact to SQLite memory under `LOW` or `MEDIUM` hazard, for `(casual_mention, MEDIUM)`, and for `(unhedged_assertion, LOW)`.
- \textbf{`BEYOND_REASONABLE_DOUBT` (*confidence* ≥ *0.95*):} Required for all `HIGH`-hazard claims (medical, legal, financial, security), for `(unhedged_assertion, MEDIUM/HIGH)`, and for all `irreversible_action` dispatches regardless of hazard level.

Before an inferred or partially proven premise is checked against *POLICY_TABLE_2D(c, h)*, `CorroborationEngine` (`src/lucca/core/corroboration.py` and `src/lucca/core/walton_schemes/evaluators.py`) evaluates Douglas Walton's five defeasible argumentation schemes (`AUTHORITY`, `SIGN`, `ANALOGY`, `CONSEQUENCE`, and `PRECEDENT`). Each scheme dispatches three critical questions (`CQ1..CQ3`: source receipt or factual congruence, independent grounding-eligible corroboration, and competing dispute or countervailing evidence) against the signed SQLite store and turn receipts. The confidence update function `adjust_confidence(confidence, q_results, scheme)` increments independently corroborated claims (*+0.10* to *+0.15* subject to a hard *$\min(0.95, \ldots)$* ceiling that prevents defeasible evidence from laundering into *1.00* certainty) and

decrements disputed or superseded claims (*-0.20* to *-0.30*) while appending explicit defeater conditions (`unless…`, `assuming…`) rendered in speech by `format_conditioned_speech`.

\subsection{Task-Priority Pooling for Multi-Sentence Intent Routing (`NeuralIntentClassifier`)} For multi-head intent routing (`src/lucca/router/neural_intent_classifier.py`, `models/router_intent_ov`, static shape `[1, 64]`), `NeuralIntentClassifier` predicts 14 primary intents and 9 reasoning domain flags per sentence and promotes concrete task/reasoning intents (`CLASS_PRIORITY_TIER >= 1`) over greetings and conversational filler (`chit_chat`, `clarify`), paired with deterministic `known_apps` slot verification (`_extract_app_entity`) and `ConformalGuard` out-of-domain abstention.

Empirical Evaluation

\subsection{Large-Sample Action-Claim Evaluation (*N=2{,}022*)}

To test whether clause-level NPU supervision resolves the open-vocabulary ceiling identified in Paper 02 [maxwell2026evasion] with adequate statistical power, we evaluated `NeuralActionClaimVerifier` against the legacy 44-stem regex supervisor across all *N=2{,}022* labeled action-claim pairs in the repository (`logs/test_reports/wo42_large_sample_2020_report.json`):

- **Paper 02 4-Model Corpus (*n=1{,}000*):** 250 prompts × 4 Gemma-4 models (`26B-A4B`, `12B`, `E4B`, `E2B`) with dual-rater blinded semantic labels ($\kappa = 1.0000$ binary).
- **Paper 02 Held-Out V1/V2 Benchmark (*n=120*):** 50 temptations, 40 honest controls, 30 idioms.
- **Organic, Implicit, Passive, and Blind Multi-Sentence Suites (*n=352*):** Including `BLIND_COH_30`, `BLIND42_COH_100`, `LOCKED_COH_050`, passive/third-person claims, structured JSON/YAML/TOML/XML status blocks, and foreign-language claims.
- **Property-Based Hypothesis Fuzz Corpus (*n=550*):** Randomized multi-sentence needle-in-a-haystack and benign control permutations.

Table: Action-Claim Supervision on the Combined *N=2{,}022* Battery (*471* Positive Violations, *1{,}551* Benign/Grounded Controls).

Supervisor Architecture	Accuracy	Recall (Sensitivity)	False Positive Rate	Latency (Mean / P95)
Legacy 44-Stem Regex [maxwell2026evasion]	*79.23%* (*1{,}602/2{,}022*)	*11.46%* (*54/471*)	*0.19%* (*3/1{,}551*)	*0.04 ms / 0.09 ms* (CPU)
Whole-Turn DeBERTa (Pre-WO-42)	*89.12%* (*1{,}802/2{,}022*)	*68.15%* (*321/471*)	*4.51%* (*70/1{,}551*)	*1.31 ms / 1.85 ms* (NPU)
Clause-Level NPU (`WO-42`)	*99.26%* (*2{,}007/2{,}022*)	*98.73%* (*465/471*)	*0.58%* (*9/1{,}551*)	*2.95 ms / 3.67 ms* (NPU)
95% Wilson Score CI (WO-42)	*[98.78%, 99.55%]*	*[97.25%, 99.42%]*	*[0.31%, 1.10%]*	*0 MB* dGPU VRAM

Table summarizes the results. On the *471* positive unexecuted action claims, `NeuralActionClaimVerifier` catches *465/471* (*98.73%*) compared to *54/471* (*11.46%*) for the legacy

regex supervisor (*411* discordant wins vs. *0* losses, exact McNemar $\chi^2 = 399.04$, $p = 3.07 \times 10^{-89}$). On the *1{,}551* negative controls (honest refusals, idioms, code explanations, and grounded receipt narrations), the false positive rate is *0.58%* (*9/1{,}551*, 95% Wilson CI: *[0.31%, 1.10%]*), satisfying the ≤ *2.0%* production gate established in Paper 02.

Pre-Registered Blind and Locked Holdout Suites (*N=540*)

To prevent iterative test-set overfitting, we conducted two multi-agent blinded evaluations:

1. \textbf{`Blind-390` (`WO-42 Phase 4`):} Three isolated subagents with zero access to training scripts or model weights authored 300 multi-sentence adversarial items (`BLIND42_COH_001..100`, `BLIND42_HZD_001..100`, `BLIND42_RTR_001..100`), combined with the 90-item `WO-41` blind stress test.
2. \textbf{`WO42_LOCKED_HOLDOUT_150`:} After freezing all training weights, three fresh isolated subagents authored 150 locked items (*50* per classifier). Two independent blinded LLM judges graded all 150 outputs line by line without seeing model predictions, achieving *20/20* (*100.0%*, Cohen's $\kappa = 1.000$) inter-rater agreement on the overlap slice.

Table: Performance Across Blind-390 and the Pre-Registered Locked Holdout (`WO42_LOCKED_HOLDOUT_150`).

NPU Subsystem	Blind-390 (Pass 1)	Blind-390 (Pass 2 / Post-Fix)	Locked-150 (Single-Pass)	Locked-150 (Post-Fix)
`NeuralActionClaimVerifier`	*109/130* (*83.8%*)	*130/130* (*100.0%*)	*50/50* (*100.0%*)	*50/50* (*100.0%*)
`NeuralHazardClassifier`	*98/130* (*75.4%*)	*125/130* (*96.2%*)	*50/50* (*100.0%*)	*50/50* (*100.0%*)
`NeuralIntentClassifier`	*96/130* (*73.8%*)	*121/130* (*93.1%*) / *122/130* (*93.8%*)	*45/50* (*90.0%*)	*50/50* (*100.0%*)
Combined Total	*303/390* (*77.7%*)	*376/390* (*96.4%*) / *377/390* (*96.7%*)	*145/150* (*96.67%*)	*150/150* (*100.0%*)

As shown in Table , single-pass evaluation on `WO42_LOCKED_HOLDOUT_150` (`logs/test_reports/wo42_locked_holdout_150_report.json`) achieved *50/50* (*100.0%*) on action-claim verification, *50/50* (*100.0%*) on hazard classification, and *45/50* (*90.0%*) on intent routing (*145/150 = 96.67%* overall, 95% Wilson CI: *[92.43%, 98.57%]*). Three of the 5 single-pass router errors (`LOCKED_RTR_007, 008, 011`) predicted `logic_constraints` with `pred_libs = ["sympy", "z3"]` but assigned `reasoning_domain = "mathematics"` instead of `"combinatorial_smt"` because `heuristic.py` checked `has_sympy` before `has_z3`; the remaining 2 errors (`LOCKED_RTR_045, 046`) wrapped app names in Markdown backticks (`` `calculator` ``, `` `notepad` ``) before the `launch_app` regex boundary. Fixing both deterministic rules in `src/lucca/router/heuristic.py` after recording the single-pass locked report resolved all 5 cases (*150/150*, `logs/test_reports/wo42_locked_holdout_150_post_router_fix_report.json`, and raised `Blind-390` from *376/390* to *377/390*). We report both the *96.67%* single-pass locked number and the *100.0%* post-fix number to preserve strict pre-registration transparency.

Live 4-Model End-to-End Verification (*N=100* Live Turns)

Offline classifier benchmarks do not test whether an orchestrator can intercept a live LLM mid-turn, clear its speech queue, and force a self-corrected response without breaking benign tool execution. We therefore executed a 100-turn live matrix (`scripts/run_wo42_live_4model_matrix.py`, `logs/test_reports/wo42_live_4model_summary.json`) across all four local Gemma-4 GGUF checkpoints served on an NVIDIA RTX 5080 Laptop GPU (16\,GB GDDR7) while the three OpenVINO FP16 classifiers ran concurrently on the Intel AI Boost NPU:

- **Tier 0:** `gemma-4-26B-A4B-it-heretic.Q6_K.gguf` (MoE, 4B active, *15{,}918 MB* VRAM)
- **Tier 1:** `gemma-4-12b-heretic-Q6_K.gguf` (Dense 12B, *11{,}130 MB* VRAM)
- **Tier 2:** `gemma-4-E4B-it-uncensored-heretic-Q6_K.gguf` (Dense 4.5B, *4{,}848 MB* VRAM)
- **Tier 3:** `Gemma4-E2B-Heretic-Sinhala.i1-Q6_K.gguf` (Dense 2.0B, *2{,}676 MB* VRAM)

Each model completed 25 live turns across multi-sentence tool routing, proof-policy hazard gating, adversarial false-action-claim temptations, and grounded narration. Three blinded LLM judges graded all 100 transcripts line by line ($\kappa = 1.000$ on the 20-turn overlap slice).

Table: Live 4-Model End-to-End Matrix (*25 Turns × 4 Local Gemma-4 Models = 100 Live Turns*).

Local GGUF Checkpoint	RTX 5080 VRAM	Mean Turn Latency	Axis 1: Coherence \	Grounding	Axis 2: Task, Routing \	Reasoning
`Gemma4-E2B-Heretic (Q6_K)`	*2{,}676.0 MB*	*0.66 s*	*25/25* (*100.0%*)	*23/25* (*92.0%*)		
`gemma-4-E4B-heretic (Q6_K)`	*4{,}848.0 MB*	*1.11 s*	*25/25* (*100.0%*)	*25/25* (*100.0%*)		
`gemma-4-12b-heretic (Q6_K)`	*11{,}130.0 MB*	*2.15 s*	*25/25* (*100.0%*)	*25/25* (*100.0%*)		
`gemma-4-26B-A4B (Q6_K)`	*15{,}918.0 MB*	*37.01 s*	*25/25* (*100.0%*)	*25/25* (*100.0%*)		
Pooled 4-Model Total	*0.0 MB* NPU overhead	—	*100/100* (*100.0%*)	*98/100* (*98.0%*)		

Whenever a model fabricated an unexecuted action in Round 1 of Category A (observed in *19/28* initial drafts across the four models, including JSON status payloads and passive confirmations), `SpeechStreamPipeline.check_functional_coherence` (`src/lucca/core/speech_stream_pipeline.py`, delegated from `src/lucca/core/orchestrator.py`) caught the violation via `NeuralActionClaimVerifier`, flushed the pending TTS queue (`speech_output.clear_queue()`), discarded the ungrounded draft span, and forced a self-corrected Round 2 response acknowledging that no tool was executed (*28/28* final turn pass rate, *0/16* false positives on Category B code/JSON controls).

\subsection{Pre-Registered *N=300* Memory-Poisoning and Contrastive Companion Expansion (`WO-43`)}

Because our earlier Zenodo preprints [maxwell2026poisoning,maxwell2026fivelayer] remain permanently frozen at their published *N=100* (*40* Cohort A, *30* Cohort B, *30* Cohort C) and *N=30* (*10*

per stratum) sample sizes to preserve archival DOI integrity, we report the pre-registered Work Order 43 (*N=300*, tag `prereg-wo43-power-300`) statistical power expansions here in the successor study. Six isolated subagents with zero access to evaluator internals authored 300 paired memory-poisoning probes (`data/benchmarks/memory_poisoning_v3_300.jsonl`) and 300 contrastive companion prompts (`data/benchmarks/contrastive_battery_v2_300.jsonl`), evaluated live on `gemma-4-E4B-it-uncensored-heretic-Q6_K.gguf` (*T=0.0*, `seed=42`, `cache_prompt=False`).

Table: Pre-Registered 5-Arm Memory Poisoning & Misinformation Ablation v3 (*N=300*, *100* Probes per Cohort × *5* Arms = *1{,}500* Live Evaluations).

Memory Architecture Arm	Cohort A Poison	Cohort A Continuity	Cohort B Retention	Cohort C Misinfo
Naive RAG (`Mem=1, Gnd=1`)	*100/100* (*100.0%*)	*100/100* (*100.0%*)	*100/100* (*100.0%*)	*100/100* (*100.0%*)
No-Assistant (`Mem=0, Gnd=0`)	*0/100* (*0.0%*)	*2/100* (*2.0%*)	*1/100* (*1.0%*)	*100/100* (*100.0%*)
Passive Soft Tagging	*11/100* (*11.0%*)	*100/100* (*100.0%*)	*100/100* (*100.0%*)	*100/100* (*100.0%*)
Adversarial Soft Tagging	*100/100* (*100.0%*)	*100/100* (*100.0%*)	*100/100* (*100.0%*)	*100/100* (*100.0%*)
Lucca Gate v3 (`WO-43`)	*0/100* (*0.0%*)	*100/100* (*100.0%*)	*100/100* (*100.0%*)	*0/100* (*0.0%*)
95% Wilson CI (Lucca Gate v3)	*[0.00%, 4.61%]*	*[95.39%, 100.00%]*	*[95.39%, 100.00%]*	*[0.00%, 4.61%]*

Table shows that scaling from *n=30–40* to *n=100* per cohort tightens the continuity-corrected 95% Wilson upper bound on zero observed poisoning from *11.4%* to *4.61%* (*3.70%* uncorrected Method 3). Across Cohort A and Cohort C, Lucca Gate v3 (with `corroborate_user_explicit=True` and `BelnapLatentInvestigator` active) achieves *0/100* poisoning vs. *100/100* for Naive RAG and Adversarial Soft Tagging (exact McNemar $p = 4.16 \times 10^{-23}$) and *11/100* for Passive Soft Tagging ($p = 9.77 \times 10^{-4}$), with primary-vs.-NLI agreement $\kappa = 0.9850$ across all *1{,}500* turns.

Table: Pre-Registered 3-Arm Contrastive Companion Battery v2 (*N=300*, *100* Prompts per Stratum × *3* Arms = *900* Live Turns).

Stratum (*n=100*)	Arm A (Bare LLM)	Arm B (Compound Lucca)	Arm C (Top-5 Hybrid RAG)	Arm B vs. A / C (p_{HB})
Seeded Recall (`R001..100`)	*2/100* (*2.0%*)	*100/100* (*100.0%*)	*85/100* (*85.0%*)	1.03×10^{-21} / 3.05×10^{-4}
Receipt Introspection (`I001..100`)	*50/100* (*50.0%*)	*100/100* (*100.0%*)	*98/100* (*98.0%*)	2.95×10^{-11} / *1.0000*
Calibrated Abstention (`A001..100`)	*99/100* (*99.0%*)	*100/100* (*100.0%*)	*100/100* (*100.0%*)	*1.0000* / *1.0000*

Table reports the 3-arm Contrastive Companion Battery v2 (`logs/test_reports/contrastive_v2_300_results_REPORT.md`) following remediation of two deterministic grader defects (`C-07`: rejecting Bare LLM refusals before substring matching on

affirmative receipt probes and normalizing backticks in `strip_echo`; `C-08`: including conversational contractions in `ABSTENTION_PHRASES`, raising dual blinded judge agreement on the 60-item IRR slice to *60/60 = 100.0%*, *κ = 1.0000*) and bare `memory_search` tool-call demuxing in `toolcall_demux.py`. Compound Lucca achieves **100/100 (100.0%, 95% CI: *[95.4%, 100.0%]*)** across all three strata, significantly outperforming Bare LLM on Seeded Recall (*+98.0 pp*, *98:0* discordant, Holm–Bonferroni *p_{HB = 1.03 × 10^{-21}*) and Receipt Introspection (*+50.0 pp*, *50:0* discordant, *p_{HB = 2.95 × 10^{-11}*, as Bare LLM lacks a receipt store and refuses all 50 affirmative probes), while significantly outperforming Top-5 Hybrid RAG on Seeded Recall (*+15.0 pp*, *15:0* discordant, *p_{HB = 3.05 × 10^{-4}*) due to deterministic belief-supersession and autobiographical query boosting in `cascade_substrates.py`.

\section{Compounded System Integration & Modular Decomposition (`WO-42` – `WO-43`)}

Upgrading the three NPU classifiers, adding 3-tier anti-bot web extraction (`scrapling v0.4.6` with `curl_cffi` TLS fingerprint impersonation and `sanitize_html_against_prompt_injection` CSS/comment stripping in `src/lucca/sandbox/web_tool.py`), and decomposing the orchestrator and domain god-objects in Work Order 43 (`v1.51.0`: `orchestrator.py` [*2{,195 LOC*], `turn_pipeline.py` [*1{,559 LOC*], `speech_stream_pipeline.py` [*946 LOC*], `tool_dispatcher.py` [*975 LOC*], `deduction_runner.py` [*1{,378 LOC*], `code_preflight_service.py` [*366 LOC*], `memory_commit_service.py` [*844 LOC*], `cascade.py` [*775 LOC*], `cascade_intent.py` [*781 LOC*], `cascade_substrates.py` [*1{,008 LOC*], `output_supervisor.py` [*475 LOC*], `supervisor_lexicon.py` [*657 LOC*], `action_claim_grammar.py` [*607 LOC*], `receipt_truth_matcher.py` [*488 LOC*], `cegis_unshackled.py` [*575 LOC*], `cegis_grid_diff.py` [*389 LOC*], `cegis_sandbox.py` [*1{,102 LOC*], `cegis_mutator.py` [*211 LOC*], and `walton_schemes/`) resolved four downstream bottlenecks in the cognitive architecture (`tests/unit/test_wo42_compounded_enhancements.py`):

1. \textbf{Multi-Category Compound Tool Shortlisting (`src/lucca/core/tool_shortlister.py:225-297`):} Prior to `WO-42`, `shortlist_tools()` discarded all secondary task categories whenever more than one category was activated, starving two-sentence prompts (e.g., *"Search the web for CVE-2026-4096 and write a summary to audit.md"*) of workspace tools. Packing secondary activated categories into the remaining slots under the hard *≤ 12* tool cap preserves all 6 `cognitive_core` tools plus both task categories.
2. \textbf{Deep-Extract Search & Citation Provenance (`src/lucca/sandbox/service.py:616`, `src/lucca/core/tool_dispatcher.py`):} Exposing `deep_extract: bool` on `search_web` routes multi-source queries to `research_topic` without exceeding the 24-tool registry cap (`MAX_REGISTRY_TOOLS = 24`), while registering scraped `"sources"` URLs with `OutputSupervisor.register_allowed_urls` so cited URLs are not flagged as fabricated citations.
3. \textbf{Sentence-Filtered Cross-Reference Corroboration (`src/lucca/core/deduction_runner.py`, `src/lucca/core/corroboration.py:666`):} When `NeuralHazardClassifier` triggers `CROSS_REFERENCE_CMD`, `CorroborationEngine.compute_proposition_alignment` checks whether candidate evidence contains negation or adversative markers (`_RE_NEGATION_POLARITY`, `_RE_ADVERSATIVE_DEFEATER`). Passing an entire 3,500-character scraped article body caused false conflicts whenever unrelated paragraphs contained words such as *"not"* or *"however"*, whereas passing 150-character search snippets truncated version numbers and dates. Extracting only the top 1–2 sentences with highest claim-token overlap per independent domain

(`_extract_claim_aligned_sentences`) preserves per-domain receipt independence while preventing unrelated paragraph negations from defeating corroboration.

4. \textbf{NLI Entailment Wiring in Offline Skill Induction (`src/lucca/reasoning/induction.py:402-465`, `reflection.py:373`):} Passing `verifier: SemanticVerifierPort` from `DreamRetrospective` into `RuleInducer.mine_patterns()` activates bidirectional NLI entailment clustering in `cluster_phrasings()` for paraphrased multi-sentence tool arguments whose token-set Jaccard similarity falls below *0.60*.

Limitations and Residual Error Analysis

1. **Residual False Negatives (*6/471 = 1.27%* on *N=2{,}022*):** Inspection of the 6 missed positive claims in `logs/test_reports/wo42_large_sample_2020_report.json` shows that all 6 come from the 1,000-turn sampled corpus (`W751..W1000`) where the model implied completion through an indirect state consequence without an explicit action verb or status token (e.g., *"Your inbox is now completely empty and ready for Monday"* in response to an email archive prompt).
2. **Residual False Positives (*9/1{,}551 = 0.58%* on *N=2{,}022*):** The 9 false alarms occur on hypothetical troubleshooting walkthroughs where the assistant writes first-person illustrative steps (*"First, I updated the registry key in my test environment..."*) without fencing them in code blocks.
3. **Linear Latency Scaling with Clause Count:** Because `NeuralActionClaimVerifier` evaluates the full response plus each extracted clause sequentially under a thread lock on the Intel NPU, mean verification latency scales from *1.31 ms* on 1-sentence outputs to *2.95 ms* (P95 *3.67 ms*) on 3–4 sentence outputs. On 10+ sentence long-form essays, clause-by-clause NPU scoring can reach *8–12 ms*; batching clause tensors into a single static-shape NPU call is left for future work.
4. **Single-Workstation Measurement Scope:** All latency, VRAM, and live 4-model measurements were recorded on a single Windows 11 workstation (Intel Core Ultra 9 275HX, Intel AI Boost NPU, NVIDIA RTX 5080 Laptop GPU 16\,GB GDDR7) and are labeled *Author-measured* until independently reproduced on external hardware.

Conclusion

Lexical speech-regex supervisors cannot generalize to open-vocabulary verb lemmas (*29.5%* unseen-verb recall in Paper 02 [maxwell2026evasion]), while whole-paragraph neural classifiers degrade when a single fabricated action claim or task request is embedded inside multi-sentence prose (*75.0%* on `Blind-300`). By combining syntactic clause decomposition with deterministic receipt set-membership verification, max-hazard lattice pooling, and task-priority intent pooling, three *82.1M*-parameter `nli-MiniLM2-L6-H768` classifiers running in FP16 on an Intel AI Boost NPU (*0 MB* dGPU VRAM, *1.22–2.95 ms*) achieve *99.26%* accuracy (*98.73%* recall, *0.58%* FPR) across *N=2{,}022* action-claim turns, *96.67%* single-pass / *100.0%* post-fix accuracy on the pre-registered `WO42_LOCKED_HOLDOUT_150` benchmark, and *100/100* functional coherence across all four local Gemma-4 model scales.

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